
COVID-19 DATA ANALYSIS

Power BI Dashboard Project

1. Project Planning

1.1 Project Overview

This project focuses on analyzing COVID-19 pandemic data to uncover critical patterns, identify systemic failures, and develop actionable insights that will strengthen global pandemic preparedness. The core deliverable is an interactive Power BI dashboard that translates raw data into strategic decision-making tools.

Project Title	COVID-19 Data Analysis Dashboard
Objective	Analyze pandemic data and build a Power BI dashboard for insights and future preparedness
Tool	Microsoft Power BI (Desktop)
Data Sources	WHO(World Health Organization), Kaggle
Timeline	12 Weeks (Planning to Deployment)
Target Users	Public health officials, researchers, healthcare administrators

1.2 Project Goals

Primary Goals
• Analyze and visualize COVID-19 spread, mortality, and recovery rates across countries and time periods
• Identify systemic problems that amplified the pandemic impact (healthcare gaps, delayed responses, misinformation)
• Build an interactive Power BI dashboard that presents findings accessibly to decision-makers
• Develop data-driven recommendations to prevent repeating past mistakes in future pandemics

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- Create a reusable analytics framework applicable to future public health emergencies

1.3 Project Phases & Timeline

Phase	Activities	Duration	Deliverable
Phase 1: Planning	Scope definition, team setup, tools selection	Week 1-2	Project charter
Phase 2: Literature Review	Research review, benchmarking, gap analysis	Week 2-3	Research report
Phase 3: Requirements	Stakeholder interviews, use case definition	Week 3-4	Requirements doc
Phase 4: Data Collection	Source identification, ETL pipeline setup	Week 4-5	Clean dataset
Phase 5: Analysis	EDA, statistical analysis, insight extraction	Week 5-8	Analysis report
Phase 6: Dashboard Dev	Power BI design, visualization, DAX measures	Week 8-10	Dashboard v1
Phase 7: Testing & Review	Validation, stakeholder feedback, revisions	Week 10-11	Dashboard v2
Phase 8: Deployment	Publishing, documentation, training	Week 12	Final deliverable

1.5 Risk Management

Risk	Likelihood	Impact	Mitigation
Inconsistent data across sources	High	High	Standardize ETL, use WHO as primary source
Incomplete historical records	Medium	Medium	Document gaps, use interpolation where valid
Stakeholder scope creep	Medium	High	Formal change control process
Power BI performance issues	Low	Medium	Optimize data model, use aggregations
Data privacy concerns	Low	High	Use only publicly available, anonymized datasets

2. Literature Review

2.1 Overview of COVID-19 Pandemic Data

The COVID-19 pandemic generated an unprecedented volume of health data. From December 2019 to 2023, the virus infected over 770 million people and caused more than 6.9 million confirmed deaths globally, with actual estimates suggesting significantly higher excess mortality. This data richness presents a unique opportunity for retrospective analysis to extract lessons for future preparedness.

Major data initiatives emerged rapidly: the Johns Hopkins CSSE dashboard became the world's primary real-time tracking resource, WHO's global surveillance system provided standardized reporting, and Our World in Data aggregated country-level statistics into accessible, comparable formats.

2.2 Identified Problems During the COVID-19 Era

2.2.1 Healthcare System Overload

Key Findings from Research
• ICU capacity was exceeded in Italy (March 2020), New York (April 2020), and Brazil (2021), forcing triage decisions that violated standard care ethics
• Healthcare worker shortages emerged globally — the WHO estimated a pre-existing gap of 18 million workers that the pandemic exacerbated dramatically
• Supply chain failures for PPE, ventilators, and oxygen created critical care deficits in low-income countries
• Hospital-acquired infections spread rapidly due to inadequate isolation infrastructure in older facilities

2.2.2 Data Reporting Failures

Critical Data Gaps Identified in Literature
• Inconsistent case definitions: countries varied in who was tested, making cross-country comparisons unreliable in early months

- Reporting delays: weekend lulls in many countries' data created artificial waves and troughs in global tracking
- Cause of death attribution: some countries required COVID-19 as the primary cause, missing co-morbidity deaths significantly
- Testing rate disparities: low-income countries with limited testing infrastructure had systematic undercounting by factors of 10x-50x
- Age and demographic breakdowns were inconsistently reported, hindering targeted public health responses

2.2.3 Delayed Government Responses

Research consistently shows that early intervention was the strongest predictor of reduced mortality. Countries that implemented restrictions within 7 days of detecting community spread saw dramatically lower death rates than those that waited 3-4 weeks. The WHO declared a Public Health Emergency of International Concern on January 30, 2020, but many countries delayed substantive action for 6-8 additional weeks.

2.2.4 Vaccine Distribution Inequity

Vaccine Equity Failures (2021-2022)

- High-income countries vaccinated 70%+ of their populations while low-income countries remained below 10% by mid-2021
- COVAX mechanism was underfunded and unable to compete with bilateral deals between manufacturers and wealthy nations
- Cold chain requirements for mRNA vaccines made distribution to tropical, low-infrastructure regions extremely challenging
- Vaccine hesitancy, fueled by misinformation, reduced uptake even in countries with adequate supply

2.3 Benchmark Studies

Study / Source	Key Contribution
Flaxman et al. (2020) — Nature	Quantified lives saved by non-pharmaceutical interventions across Europe
Haug et al. (2020) — Nature Human Behaviour	Ranked effectiveness of 6,000+ containment measures globally
Ritchie et al. — Our World in Data	Created comprehensive comparative dataset used by 150+ countries

WHO (2021) — IPPPR Report	Identified systemic WHO and national government failures in early response
Woolf et al. (2021) — JAMA	Calculated excess mortality vs. official COVID-19 death counts for US
Robertson et al. (2020) — Lancet Global Health	Estimated indirect deaths from disrupted maternal/child health services

2.4 Research Gaps This Project Addresses

- Most existing analyses are country-specific; this project provides comparative cross-regional analysis
- Few dashboards integrate economic indicators alongside health metrics in a unified view
- Vaccination efficacy analysis rarely includes equity-adjusted metrics for low-income regions
- Real-time preparedness frameworks linked directly to historical data are absent in most tools

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder Group	Information Needs
Ministry of Health Officials	National trends, ICU capacity, mortality forecasts, vaccination coverage
Hospital Administrators	Regional hospitalization rates, resource allocation signals, surge indicators
Public Health Researchers	Granular data exports, statistical breakdowns, reproducible methodology
Policy Makers	Economic impact, intervention effectiveness, comparative country performance
General Public / Media	Simple, accessible trend visualizations, daily/weekly summaries
International Organizations (WHO, UN)	Cross-country comparability, equity metrics, preparedness benchmarks

3.2 Functional Requirements

3.2.1 Dashboard Views Required

Core Dashboard Pages
• Overview Page: Global summary KPIs — total cases, deaths, recoveries, vaccination rate
• Epidemiological Trends: Time-series charts for daily/weekly/monthly case and death counts
• Geographic Analysis: Interactive world map with drill-down to country and regional level
• Healthcare Capacity: ICU utilization, hospitalization rates, healthcare worker availability
• Vaccination Progress: Rollout speed, coverage by income group, efficacy correlation with mortality
• Response Effectiveness: Government intervention timeline vs. case/death curve analysis
• Equity Analysis: Disparities by income level, geography, age, and demographic group
• Preparedness Scorecard: Evidence-based recommendations with readiness indicators

3.3 Data Requirements

Dataset	Source	Granularity	Update Frequency
Case & Death Counts	WHO / JHU CSSE	Country / Daily	Daily
Vaccination Data	Our World in Data	Country / Daily	Daily
Testing Rates	Our World in Data	Country / Weekly	Weekly
ICU & Hospital Capacity	ECDC / National Reports	Country / Weekly	Weekly
Government Interventions	Oxford COVID-19 Gov. Response Tracker	Country / Daily	As updated
Economic Indicators	World Bank / IMF	Country / Annual	Annual
Demographic Data	UN Population Division	Country	Static
Excess Mortality	The Economist / WHO Estimates	Country / Weekly	As available

3.4 Non-Functional Requirements

Category	Requirement
Performance	Dashboard pages must load within 3 seconds; data refresh within 15 minutes
Accessibility	WCAG 2.1 AA compliant; color-blind-safe palette; screen reader compatible
Scalability	Must handle 200+ countries, 3+ years of daily data without performance degradation
Usability	Non-technical users must navigate without training; interactive tooltips throughout
Data Accuracy	All figures must be traceable to primary sources; discrepancies must be flagged
Security	Published to Power BI Service with role-based access control for sensitive views

3.5 DAX Measures Required

Key Power BI DAX Calculations
- Case Fatality Rate (CFR) = $\text{DIVIDE}([\text{Total Deaths}], [\text{Total Cases}]) * 100$ 7-Day Rolling Average Cases = $\text{AVERAGEX}(\text{DATESINPERIOD}(\text{Date}[\text{Date}], \text{LASTDATE}(\text{Date}[\text{Date}]), -7, \text{DAY}), [\text{Daily Cases}])$ - Vaccination Coverage % = $\text{DIVIDE}([\text{Total Vaccinated}], [\text{Population}]) * 100$

- Excess Mortality Rate = $\text{DIVIDE}([\text{Observed Deaths}] - [\text{Expected Deaths}], [\text{Expected Deaths}]) * 100$

- Response Delay Index = Days between first case detection and first containment measure

- Healthcare Strain Index = $\text{DIVIDE}([\text{COVID Hospitalizations}], [\text{ICU Capacity}])$

- Equity-Adjusted Vaccination Score = Vaccination rate weighted by income group disparity

4. System Analysis

4.1 Current State Analysis: What Went Wrong

The COVID-19 pandemic exposed severe structural vulnerabilities across health systems, data infrastructure, and policy coordination mechanisms. This section analyzes the root causes of failures, quantifies their impact, and translates findings into dashboard design decisions.

4.2 Root Cause Analysis of Major Problems

4.2.1 Healthcare System Failures

Problem Identified	Root Cause & Evidence
ICU overflow in multiple countries	Decades of underfunding: average OECD ICU beds were 12/100k; Italy had 8, was overwhelmed at 15/100k COVID patients
PPE shortages globally	Just-in-time supply chains with no strategic reserves; 94% of N95 masks produced in China caused global dependency
Healthcare worker burnout and death	Inadequate staffing ratios + absence of psychological support infrastructure
Delayed treatment protocols	Lack of pre-established pandemic clinical protocols; physicians improvised without standardized guidance

4.2.2 Data Infrastructure Failures

Data Problem	System Cause
Inconsistent case definitions	No pre-agreed international surveillance standard activated at pandemic onset
Weekend reporting gaps	Manual data entry processes in most national health systems; no automation
Undercounting in low-income countries	Absence of standardized vital registration systems and diagnostic capacity
Siloed data systems	National health databases were incompatible; data-sharing MoUs not in place pre-pandemic

4.2.3 Policy Response Failures

Evidence-Based Policy Failures

- Border closure delays: Average of 42 days passed between WHO PHEIC declaration and international travel restrictions
- Lockdown timing: Countries that locked down after 100 deaths/million had 3x higher peak mortality than those that acted at 10 deaths/million
- School closures: Inconsistent, politically driven decisions rather than epidemiologically justified — created childcare crises without clear benefit
- Communication failures: Mixed messaging on masks (WHO initially discouraged public mask use) eroded public trust
- Misinformation void: Government communication vacuums were filled by social media misinformation, reducing compliance

4.3 Proposed Dashboard Solution

The Power BI dashboard directly addresses the identified failures by providing a unified, evidence-based analytical platform. The system architecture is designed to transform raw pandemic data into actionable intelligence.

4.3.1 Data Architecture

Power BI Data Model Structure

- Fact Tables: DailyCovidStats (cases, deaths, recoveries, tests per country per day)
- Fact Tables: VaccinationStats (doses administered, people vaccinated, booster coverage)
- Fact Tables: HospitalizationStats (ICU occupancy, hospital admissions, healthcare worker cases)
- Dimension Tables: DimCountry (country metadata, region, income group, population, healthcare capacity)
- Dimension Tables: DimDate (full date table with week, month, quarter, pandemic phase markers)
- Dimension Tables: DimIntervention (government measures, lockdown levels, travel restrictions)
- Bridge Tables: CountryInterventionBridge (linking interventions to countries with start/end dates)

4.4 Insights to Highlight on Dashboard

4.4.1 Problems the Dashboard Visualizes

Dashboard Insight Panels

- INSIGHT 1 — Healthcare Capacity Crisis: ICU utilization map showing which countries exceeded capacity and when, correlated with mortality spikes
- INSIGHT 2 — Response Speed vs. Outcomes: Scatter plot showing countries' days-to-lockdown vs. total deaths per million — clear inverse correlation
- INSIGHT 3 — Testing Gaps & Hidden Spread: Test positivity rates revealing true pandemic scale beyond confirmed cases
- INSIGHT 4 — Vaccine Equity Divide: Side-by-side comparison of vaccination rates in high vs. low income countries with mortality correlation
- INSIGHT 5 — Excess Mortality Truth: Comparison of official COVID deaths vs. excess mortality estimates by country
- INSIGHT 6 — Economic Cost Analysis: GDP impact vs. stringency index to show cost-benefit of intervention strategies

4.5 Recommendations for Future Pandemic Preparedness

Based on the data analysis embedded in the dashboard, the following evidence-based recommendations are generated as the primary output of this project:

Preparedness Recommendations (If Pandemic Recurs)

- READINESS-1: Maintain national strategic PPE and medical supply reserves equivalent to 90-day surge demand — never rely on single-source supply chains
 - READINESS-2: Pre-define legal frameworks for emergency powers, enabling rapid activation within 48 hours of PHEIC declaration rather than weeks
 - READINESS-3: Implement real-time digital disease surveillance integrated with WHO global system — eliminate manual weekend reporting gaps
 - READINESS-4: Establish pre-negotiated international data-sharing agreements for health data before emergencies occur
 - READINESS-5: Increase ICU baseline capacity by 40% in health systems currently below OECD median (critical buffer for surge)
 - READINESS-6: Create regional vaccine manufacturing hubs in Africa, Southeast Asia, and Latin America to prevent distribution inequity
 - READINESS-7: Develop and pre-test standardized public communication protocols to counter misinformation with clear, consistent messaging
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- READINESS-8: Establish healthcare worker reserve corps with standardized training for rapid redeployment during surge events
- READINESS-9: Build pandemic impact economic models in advance to enable rapid fiscal response decisions without political delay
- READINESS-10: Mandate annual pandemic preparedness exercises (tabletop + operational) for all national health ministries

4.6 Dashboard KPI Framework

KPI Category	Metric	Target (Future Pandemic)	Dashboard Alert Level
Healthcare Capacity	ICU Occupancy Rate	< 70%	Red > 85%, Yellow 70-85%
Epidemic Control	Effective Reproduction Number (Rt)	< 1.0	Red > 1.2, Yellow 1.0-1.2
Response Speed	Days from Detection to Action	< 7 days	Red > 21 days
Testing Coverage	Test Positivity Rate	< 5%	Red > 10%, Yellow 5-10%
Vaccination	Coverage of Vulnerable Groups	> 80%	Red < 40%, Yellow 40-70%
Data Quality	Reporting Completeness Score	> 95%	Red < 80%, Yellow 80-95%

4.7 Technical Implementation — Power BI

Component	Implementation Approach
Data Ingestion	Power Query M scripts connecting to CSV exports from WHO/OurWorldInData APIs
Data Transformation	Standardized country code mapping (ISO 3166-1 alpha-3), date normalization, null handling
Data Model	Star schema with fact-dimension separation; relationships managed via country code and date keys
Visualizations	Filled maps, line charts, scatter plots, bar charts, KPI cards, matrix tables, waterfall charts
Interactivity	Cross-filtering, drill-through pages, slicer panels (country, date range, region, income group)
DAX Calculations	Time intelligence functions for rolling averages, YoY comparisons, cumulative totals

Refresh Schedule	Automated daily refresh via Power BI Service with data gateway
Distribution	Published to Power BI Service; embedded links for public access; PDF snapshots for reports
